



Department of Electrical and Electronics Engineering
Academic Year 2024 - 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE

Course Code & Title: EE3503 Control Systems

Name of the Faculty member : Dr.B.Deepa Lakshmi

Innovative Practice Description

- **Unit / Topic:** Unit I / Open loop and Closed loop system
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 1
- **Activity Chosen:** One Minute Paper
- **Justification:**

The topic open loop and closed loop system is chosen because this topic is the basic concept to analyse any system. So, one minute paper will make the students to recollect the topic and write about the concept given. After teaching the topic, I conducted this activity to make the students to understand the topic and write it in a paper.

- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation:**

After teaching the concept, I made all the students to write about open loop and closed loop control system. Students enthusiastically started writing about the topic and submit. This enhances the students to recollect and write it immediately which will be remembered by them.

- **CO – PO / PSO mapping:**

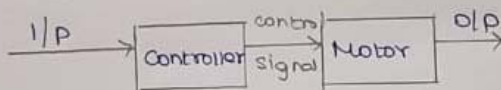
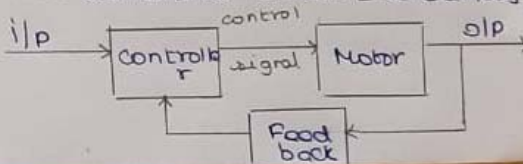
CO	PO 1	PO 2	PO 3	PO4	PO5	PO 8	PO 10	PO 12	PSO 1	PSO 3
CO1	3	3	1	2	2	1	1	1	2	2

- **PO / PSO mapped:**

Innovative practice	PO 10
	2
Justification for correlation	The student will be able to explain about open loop and closed loop control system on their own observance during teaching.



- Screenshot of the activity:

Open-loop control system	closed loop control system
No feedback is given to the control system	A feedback is given to the control system
small bandwidth	large bandwidth
More stable	Less stable
simple design	complex design
cheap	Costly
Very sensitive in nature	Less sensitive to disturbances
	

Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**

Students felt that writing about the given topic make them understand it well and able to recollect immediately.

- ❖ **Benefit of the practice:**

1. All the students were able to write the given topic.
2. After conducting the activity, I came to know that students understood the topic and they were able to explain it.

- ❖ **Challenges faced in implementation:**

1. Time utilization for conducting the activity
2. Some students could not be able to recollect and write immediately.

References:

- ✓ J.Nagrath and M.Gopal, "Control Systems Engineering", New Age International Publishers



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Innovative Practice Description

- **Unit / Topic:** Unit II / Time response
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 4
- **Activity Chosen:** MATLAB Simulation Tool
- **Justification:**

The topic is chosen because students are studying the Time response theoretically, made them to apply and plot the time response of the given system. So, I took all the students to the lab and make them use individual system to learn how to plot time response.

- **Time Allotted for the Activity:** 30 minutes

- **Details of the Implementation:**

After teaching the concept, I took the students to the laboratory and provide a single system for everyone to use MATLAB software. Then I instructed them to obtain time response for the given system. Students eagerly write the coding and simulate it. All the students could able to obtain the response. This enhances the students to know a new simulation tool for analysis.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 8	PO 10	PO 12	PSO 1	PSO 3
CO2	3	3	1	2	2	1	1	1	2	2

- **Screenshot of the activity:**



Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

Students felt that knowing the new tool is interesting and correlating the plot obtained theoretically with the MATLAB output is enhancing their knowledge to compare the concepts.

❖ ***Benefit of the practice:***

1. All the students were able to use the MATLAB software.
2. After conducting the activity, I came to know that students could be able to plot the time response for any given system.

❖ ***Challenges faced in implementation:***

1. Teaching a new tool in half an hour is difficult

References:

- ✓ J.Nagrath and M.Gopal, “Control Systems Engineering”, New Age International Publishers



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Innovative Practice Description

- **Unit / Topic:** Unit III / Bode Plot
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 7
- **Activity Chosen:** MATLAB Simulation Tool
- **Justification:**

The topic is chosen because students are studying the frequency response using bode plot theoretically, made them to apply and plot the frequency response of the given system. So, I took all the students to the lab and make them use individual system to learn how to plot frequency response.

- **Time Allotted for the Activity:** 30 minutes

- **Details of the Implementation:**

After teaching the concept, I took the students to the laboratory and provide a single system for everyone to use MATLAB software. Then I instructed them to obtain frequency response using bode plot for the given system. Students eagerly write the coding and simulate it. All the students could able to obtain the response. This enhances the students to know a new simulation tool for analysis.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 8	PO 10	PO 12	PSO 1	PSO 3
CO3	3	3	1	2	2	1	1	1	2	2

- **Screenshot of the activity:**



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students felt that knowing the new tool is interesting and correlating the plot obtained theoretically with the MATLAB output is enhancing their knowledge to compare the concepts.

❖ *Benefit of the practice:*

1. All the students were able to use the MATLAB software.
2. After conducting the activity, I came to know that students could be able to plot the time response for any given system.

❖ *Challenges faced in implementation:*

1. Teaching a new tool in half an hour is difficult

References:

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Innovative Practice Description

- **Unit / Topic:** Unit IV / Controllability and Observability
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO - 12
- **Activity Chosen:** Exit Slips
- **Justification:**

The topic controllability and observability is chosen because the topic is important to control the states of the system. So, exit slips will make the students to recollect the topic and write about the concept given. After teaching the topic, I conducted this activity to make the students to understand the topic and write it in a paper.

- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation:**

After teaching the concept, I made all the students to write about controllability and observability. Students enthusiastically started writing about the topic and submit. This enhances the students to recollect and write it immediately which will be remembered by them.

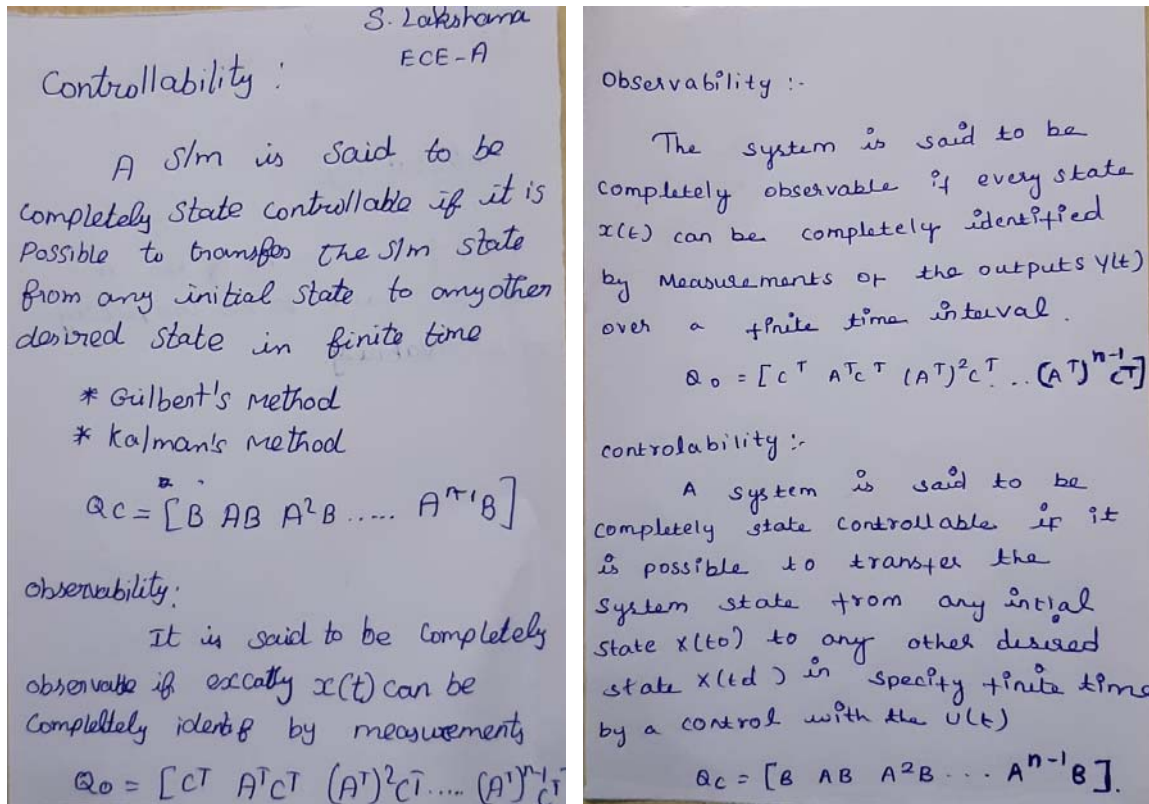
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO4	PO5	PO 8	PO 10	PO 12	PSO 1	PSO 3
CO4	3	3	1	2	2	1	1	1	2	2

- **PO / PSO mapped:**

Innovative practice	PO 2
	2
Justification for correlation	The student will analyse the controllability and observability of the system to measure states of the system whether the system is state controllable or observable.

• **Screenshot of the activity:**



Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

Students felt that writing about the given topic make them understand it well and able to recollect immediately.

❖ **Benefit of the practice:**

1. All the students were able to write the given topic.
2. After conducting the activity, I came to know that students understood the topic and they were able to explain it.

❖ **Challenges faced in implementation:**

1. Time utilization for conducting the activity
2. Some students could not be able to recollect and write immediately.

References:

- ✓ J.Nagrath and M.Gopal, "Control Systems Engineering", New Age International Publishers
- ✓ https://www.readingrockets.org/strategies/exit_slips